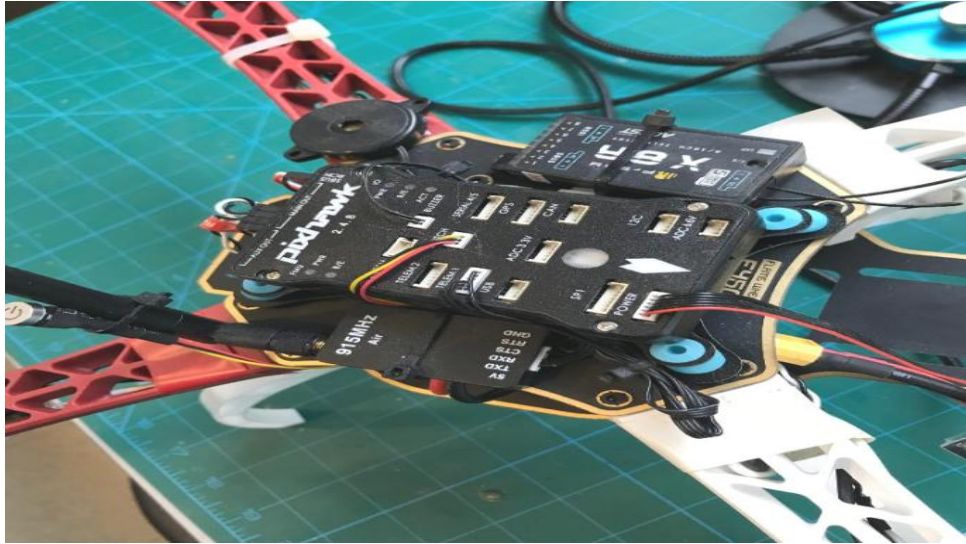


Autonomous Control Design

I am attempting to create an autonomous controller for the Flame Wheel 450 as shown below:



and carry out drone missions using QGroundControl. I am currently in the stage of creating the necessary SIMULINK model to achieve stability. I am also in the process of ordering all of the necessary hardware to make this possible. So far, I've ordered these parts:

Airspeed sensor - pitot tube
GPS -
RFD900x Long Range Telemetry Modem - connects drone data to ground station
TFmini-S Micro LiDAR Module - measures distance from ground and other objects
PixHawk Controller - controller
ADTi 20MP 20L V1 APS-C Drone Mapping Camera
Raspberry Pi - acts as a bridge between sensors/code and flight controller

And once they arrive, I will begin to assemble them onto the drone/my computer. So far this is my unfinished Simulink model utilizing PID Control.

